

Chuẩn bị Dữ liệu

Analysis: Data Preparation

Trí tuệ Nhân tạo cho Kế toán (AI in Accounting)

Khoa Kế toán - Kiểm toán

Bài giảng Thực hành 05

Góc nhìn Chuyên gia (Professional Insight)

- **Thực tế 80/20:** Chuẩn bị dữ liệu thường chiếm 80% thời gian của một dự án phân tích dữ liệu, chỉ 20% còn lại dành cho việc phân tích thực sự.
- **Rác vào - Rác ra (GIGO):** Dữ liệu không được chuẩn bị kỹ lưỡng sẽ dẫn đến kết quả phân tích sai lệch, bất kể mô hình học máy (Machine Learning) có phức tạp đến đâu.
- **Tầm quan trọng của ETL:** Quá trình Trích xuất (Extract), Chuyển đổi (Transform), và Tải (Load) là kỹ năng cốt lõi bắt buộc đối với một Kế toán viên trong kỷ nguyên số.

Lộ trình Chương (Chapter Roadmap)

- **LO 5.1:** Định hình dữ liệu (Data Profiling).
- **LO 5.2:** Mô tả quá trình ETL.
- **LO 5.3:** Mẫu trích xuất dữ liệu.
- **LO 5.4:** Mẫu chuyển đổi cột (Column Transformation).
- **LO 5.5:** Mẫu chuyển đổi bảng (Table Transformation).
- **LO 5.6:** Mẫu chuyển đổi mô hình dữ liệu.
- **LO 5.7:** Mẫu tải dữ liệu.

5.1 Quá trình Định hình Dữ liệu (Data Profiling)

- **Data Profiling là gì?** Là bước khảo sát đầu tiên để hiểu rõ dữ liệu trước khi xử lý. Giống như việc "khám sức khỏe tổng quát" cho dữ liệu.
- **Hai bước định hình chính:**
 - *Điều tra chất lượng (Investigate Quality):* Tìm các giá trị bị thiếu (Missing), bất thường (Anomalies), hoặc sai định dạng.
 - *Điều tra cấu trúc (Investigate Structure):* Hiểu cách các cột (trường) và hàng (bản ghi) được tổ chức.
- **Quyết định & Thông báo:** Khi phát hiện lỗi nghiêm trọng, cần quyết định loại bỏ/sửa chữa dữ liệu và thông báo cho các bên liên quan.

Data Preparation Process

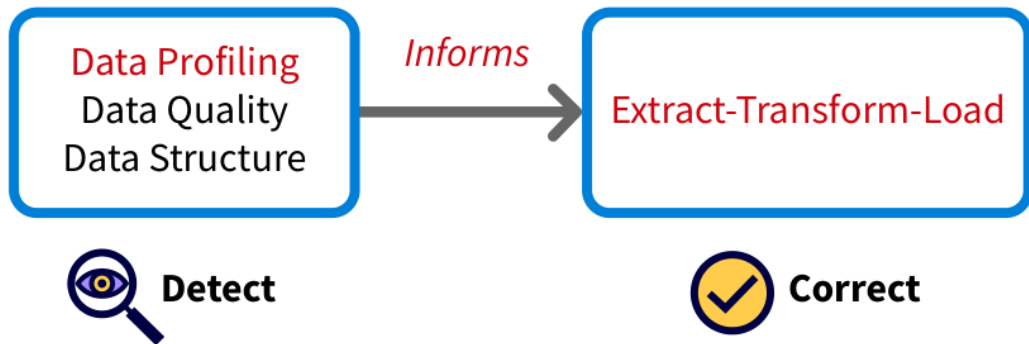


ILLUSTRATION 5.1A

Assessing Data Reliability and Impact

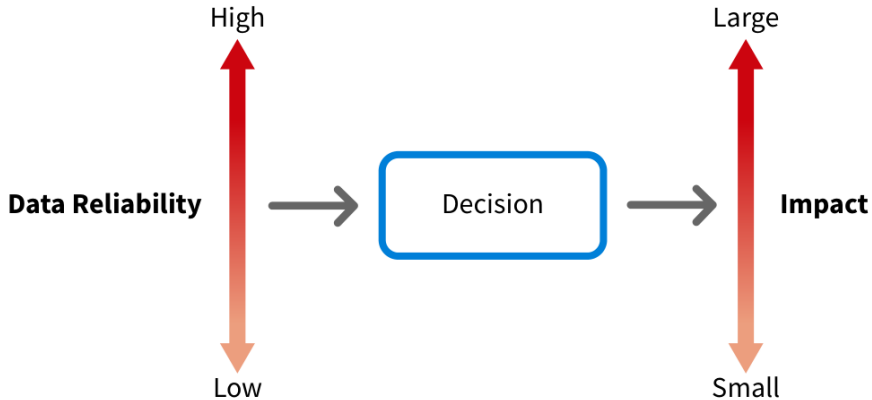


ILLUSTRATION 5.2

(A) Composite Column

Name
Cole, Lakeisha, NA
Lopez, Alejandro, CPA
Key, Kim, NA
David, Julie, CPA
Malone, Moses, NA
Buslepp, Bill, CPA
Potoms, Keme, NA
Despontin, Marc, CPA
Kaminski, Ivanka, NA

(B) Single-Valued Column

Name	Certification
Cole, Lakeisha	NA
Lopez, Alejandro	CPA
Key, Kim	NA
David, Julie	CPA
Malone, Moses	NA
Buslepp, Bill	CPA
Potoms, Keme	NA
Despontin, Marc	CPA
Kaminski, Ivanka	NA

ILLUSTRATION 5.3

(A) Multi-Valued Column

Name	Certifications
Cole, Lakeisha	CISA
Lopez, Alejandro	CPA, CISA
Key, Kim	CMA
David, Julie	CPA, CMA, CISA
Malone, Moses	CPA
Buslepp, Bill	CPA, CMA
Potoms, Keme	CMA
Despontin, Marc	CPA
Kaminski, Ivanka	CMA, CISA

(B) Single-Valued Column

Name	Certifications
Cole, Lakeisha	CISA
Lopez, Alejandro	CISA
Lopez, Alejandro	CPA
Key, Kim	CMA
David, Julie	CISA
David, Julie	CMA
David, Julie	CPA
Malone, Moses	CPA
Buslepp, Bill	CMA
Buslepp, Bill	CPA
Potoms, Keme	CMA
Despontin, Marc	CPA
Kaminski, Ivanka	CISA
Kaminski, Ivanka	CMA

ILLUSTRATION 5.4

(A) Crosstabulation

	Jim	Jane	Elodie	Carlos
Focus	0	1	5	4
Mustang	1	2	3	3
Escape	4	1	0	2
Explorer	2	2	0	1

(B) Flat

Model	SalesPerson	UnitsSold
Focus	Carlos	4
Mustang	Carlos	3
Escape	Carlos	2
Explorer	Carlos	1
Focus	Elodie	5
Mustang	Elodie	3
Escape	Elodie	0
Explorer	Elodie	0
Focus	Jane	1
Mustang	Jane	2
Escape	Jane	1
Explorer	Jane	2
Focus	Jim	0
Mustang	Jim	1
Escape	Jim	4
Explorer	Jim	2

ILLUSTRATION 5.5

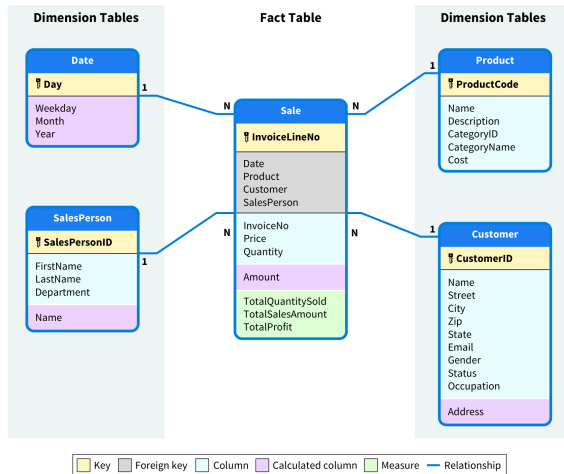


ILLUSTRATION 5.6

1

What is the **total sales amount** (dollars) for a specific **customer**?

2

What is the **total sales amount** (dollars) for all customers in a specific **state**?

3

What is the **total quantity sold** of a specific **product**?

4

Which **month** has the highest **total sales amount** (dollars)?

5

What is the **total sales amount** (dollars) for a specific **product** to a specific **customer**?

6

How do the **monthly** sales, in terms of **total quantity sold** for a specific **product**, fluctuate?

7

Who is the **salesperson** with the highest **total sales amount** (dollars) for a specific **product**?

8

What is the **total quantity sold** of a specific **product** in a specific **month** in a specific **state**?



Dimensions



Measures

ILLUSTRATION 5.7

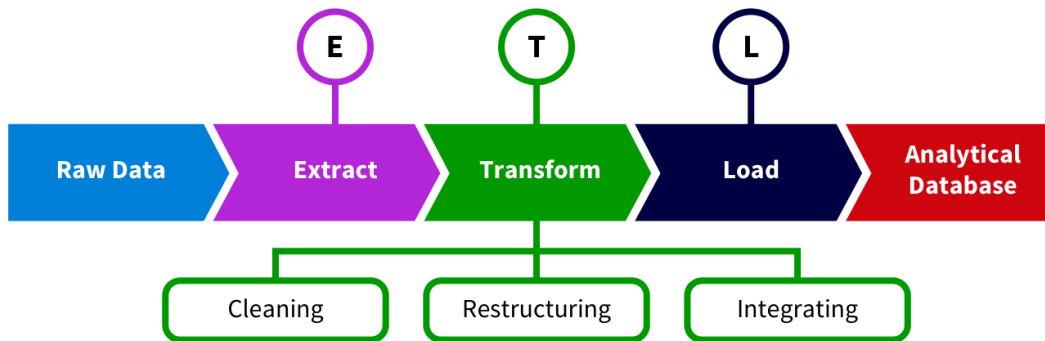
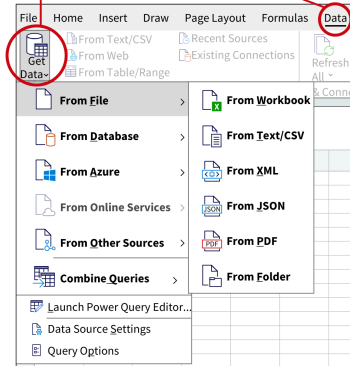


ILLUSTRATION 5.8

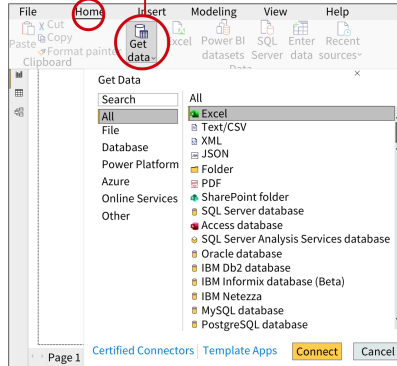
(A) Excel Data Connectors

For an overview of all Excel's data connectors, click on the Data tab in the Main Menu, then select Get Data in the Ribbon.



(B) Power BI Data Connectors

For an overview of all Power BI's data connectors, click on the Home tab in the Main Menu, then select Get Data in the Ribbon.



5.2 Mô tả quá trình ETL (Extract, Transform, Load)

- **E - Extract (Trích xuất):** Kéo dữ liệu từ các hệ thống nguồn (ERP, CRM, Web). Dữ liệu này thường ở dạng thô và chưa sẵn sàng để phân tích.
- **T - Transform (Chuyển đổi):** Là khâu quan trọng nhất. Làm sạch, tính toán lại, định dạng lại để dữ liệu phù hợp với mô hình phân tích mục tiêu.
- **L - Load (Tải):** Đưa dữ liệu đã được làm sạch vào kho dữ liệu (Data Warehouse) hoặc phần mềm trực quan hóa (Power BI, Tableau).

ILLUSTRATION 5.9

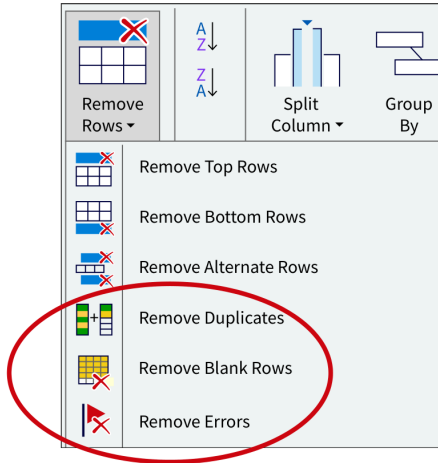


ILLUSTRATION 5.10

Replace Values

×

Replace one value with another in the selected columns.

Value To Find

M

Replace With

Male

Advanced options

OK

Cancel

ILLUSTRATION 5.11

**(A) Customer Table:
Financial Information**

Names	Status	Discount
William McCarthy	A	5
Margarita David	C	1
Ann White	B	3
Carlos Panetta	A	5
Jen Pollack	B	3

**(B) Customer Table:
Demographic Information**

Names	Gender	Age
Carlos Paretta	M	44
David Margo	M	22
Jenny Pollack	F	57
Bill McCarthy	M	38
Ann White	F	28

(C) Merged Customer Table

Names	Status	Discount	Gender	Age
Ann White	B	3	F	28
Carlos Panetta	A	5	M	44
Jen Pollack	B	3	F	57
Margarita David	C	1	M	22
William McCarthy	A	5	M	38

ILLUSTRATION 5.12

Worksheet	Description
ClData	Information about Beans' clients.
Employee	General information about Beans' employees.
E-Dem	Demographic information about Bean's employees.
Service	Information about the services provided for the period January-July 2025.

ILLUSTRATION 5.13A

ClData

Name	Description
ID	A client's unique ID.
Name	A client's name.
Industry Name	A client's industry.

Employee

Name	Description
ID	An employee's ID.
Name	An employee's name.
Job Title	An employee's job title.
Rate	An employee's hourly rate.
Office	An employee's office.
MS	An employee's marital status.

E-Dem

Name	Description
First	An employee's first name.
Last	An employee's last name.
Age	An employee's age.
CertList	List of an employee's certifications.
University	The university from which the employee received an undergraduate degree.

(Continued)

ILLUSTRATION 5.13B

Service

Name	Description
ID	A unique ID for a service provided.
Date	The date the service occurred.
Actual Time	The actual time spent on a service provided.
Budgeted Time	The budgeted time for a service provided.
ID	The ID of the task performed.
Area	The task's area.
Task	The name of the task performed.
EmployeeID	The ID of the employee performing the task.

ILLUSTRATION 5.14

Select "ID" Column

	A	B	C	D	E	F	G	H	I	J
1	ID	Date	ActualTime	BudgetedTime	ID	Area	Task	Employee	Client	
2	1	1/5/2023	3.75	3.5	2	Accounting	Review	18	42	
3	2	1/6/2023	2.65	2.5	1	Accounting	Prep	15	103	
4	3	1/6/2023	0.07	0.1	1	Accounting	Prep	12	178	
5	4	1/6/2023	0.35	0.3	1	Accounting	Prep	9	178	
6	5	1/6/2023	0.1	0.1	1	Accounting	Prep	2	2	
7	6	1/6/2023	0.35	0.4	1	Accounting	Prep	9	2	
8	7	1/6/2023	0.07	0.1	1	Accounting	Prep	12	2	
9	8	1/6/2023	0.25	0.3	1	Accounting	Prep	15	2	
10	9	1/6/2023	0.25	0.2	1	Accounting	Prep	17	2	

Workbook Statistics Caps Lock Average: 2316.5 Count: 4633 Numerical Count: 4632 Min: 1 Max: 4632 Sum: 10730028

Status Bar

Control Total: Average

Number of Rows

Control Total: Sum

ILLUSTRATION 5.15

Column Statistics	...
Count	4,632
Error	0
Empty	0
Distinct	4,632
Unique	4,632
NaN	0
Zero	0
Min	1
Max	4,632
Average	2,316.5
Standard deviation	1,337.28...
Even	2,316
Odd	2,316

Number of Rows

Control Total:
Average

ILLUSTRATION 5.16

(A) Column with Unreliable Data

Age
45
1998
thirty two
55
34
22
??
??
51
9/2/1998

(B) Removing a Column with Unreliable Data

The screenshot shows the QlikView interface with the 'Remove Columns' button circled in red. The 'ABC Age' column is also circled in red. The data table below shows the current state of the data.

	ABC First	ABC Last	ABC Age	ABC Certlist	ABC University
1	Diego	Asare	null	null	Grand Valley State University
2	Kim	Bush	45	CPA	University of Delaware
3	Marcus	Clay	1998	null	MSU
4	Gail	David	thirty two	null	UM
5	Shiela	Diamond	55	CPA, CMA, CFA	Michigan State University
6	Ed	Diamond		CPA	MSU
7	John	Federer	34	null	University of Michigan
8	Kayne	James	22	null	Michigan State University
9	Sara	Lee	??	null	Michigan State University
10	Ivan	Lenk	??	CPA	University of Michigan

5.3 Áp dụng các mẫu để Trích xuất Dữ liệu (Extraction Patterns)

- **Mẫu trích xuất toàn bộ (Full Extraction):**

- Rút toàn bộ dữ liệu từ nguồn ở mỗi lần thực hiện. Phù hợp với các bảng dữ liệu nhỏ (Ví dụ: Danh mục khách hàng).

- **Mẫu trích xuất tăng dần (Incremental Extraction):**

- Chỉ trích xuất những dữ liệu mới được tạo hoặc bị thay đổi kể từ lần trích xuất cuối cùng. Giúp tiết kiệm băng thông và tài nguyên hệ thống (Ví dụ: Dữ liệu giao dịch bán hàng hàng ngày).

ILLUSTRATION 5.17

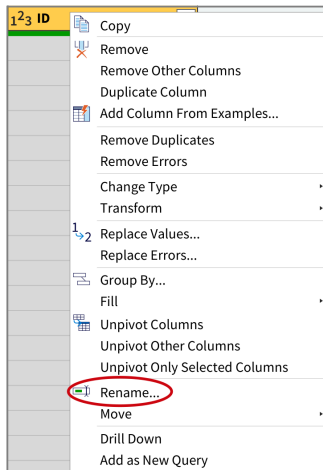


ILLUSTRATION 5.18

Table	ColumnName	RevisedColumnName
ClData	ID	ClientID
ClData	Industry Name	IndustryName
Employee	ID	EmployeeID
Employee	Job Title	JobTitle
Employee	MS	MaritalStatus
E-Dem	First	FirstName
E-Dem	Last	LastName
E-Dem	CertList	CertificationList
Service	ID	ServiceID
Service	Actual Time	ActualTime
Service	Budgeted Time	BudgetedTime
Service	ID-1	TaskID
Service	Task	TaskName

ILLUSTRATION 5.19

(A) Raw
Spreadsheet
Data

A
ID
1
2
A
4
5
B
6

(B) Extracted Data

ABC 123 ID	
	1
	2
A	
	4
	5
B	
	6

(C) Data Type Transformation:
Whole Number

1	ID
2	2 (28%)
3	Error

Remove Errors ...

1	1
2	2
3	Error
4	4
5	5
6	Error
7	6

(D) Data Type Transformation:
Text

 A ^B C ID 	
1	1
2	2
3	A
4	4
5	5
6	B
7	6

ILLUSTRATION 5.20

ABC 123 ID ▼		
1	1.2	Decimal Number
2	\$	Fixed Decimal Number
3	1 ² 3	Whole Number
4	%	Percentage
5		Date/Time
6		Date
7		Time
		Date/Time/Timezone
		Duration
	ABC	Text
		True/False
		Binary
		Using Locale...

ILLUSTRATION 5.21

The screenshot shows the Microsoft Access interface. The 'Home' tab is selected in the ribbon. The 'Split Column' button is circled in red. A context menu is open, showing options for splitting columns by delimiter, number of characters, positions, and case/digit rules. The 'Employee' table is selected in the 'Queries [4]' list. The 'EmployeeID' and 'Name' fields are highlighted in the table view.

File Home Transform Add Column View Tools Help

Close & Apply Close New Source New Query Recent Sources Enter Data Data Source Settings Data Source Manage Parameters Parameters Refresh Preview Properties Advanced Editor Manage Columns Reduce Rows Sort

Queries [4]

- CLdata
- E-DEM
- Employee
- Service

	1 ² EmployeeID	AB ^C Name	AB ^C Job
1	1	Kanya, James	Staff
2	2	Ivan, Lenk	Manager
3	3	Mick, Richards	Senior
4	4	Juan, Lozano	Staff
5	5	Gail, David	Senior

Split Column

- By Delimiter
- By Number of Characters
- By Positions
- By Lowercase to Uppercase
- By Uppercase to Lowercase
- By Digit to Non-Digit
- By Non-Digit to Digit

Data Type: Text Use First Row as Headers Replace Values

ILLUSTRATION 5.22

Split Column by Delimiter?×

Specify the delimiter used to split the text column.

Select or enter delimiter

Comma

↑

Split at

☒ Left-most delimiter

☐ Right-most delimiter

☐ Each occurrence of the delimiter

> Advanced options

ILLUSTRATION 5.23

(A) ActualTime Column: Statistics

Column Statistics	...
Count	4,632
Error	0
Empty	0
Distinct	190
Unique	48
NaN	0
Zero	0
Min	0.05
Max	25
Average	1.77672...
Standard deviation	2.03039...

**(B) ActualTime Column:
Values in Descending Order**

1 ² 3 ID ▼	1.2 ActualTime ▼
1325	25
3277	22.5
4387	20
2828	11.4
2452	10.75

ILLUSTRATION 5.24

ID	IncorrectValue	CorrectValue
1325	25	2.5
3277	22.5	2.25
4387	20	2

ILLUSTRATION 5.25

(A) JobTitle Column: Distinct Values

▼ ABC LastName ▼ ABC JobTitle ▼

Sort Ascending
Sort Descending
Clear Sort

Clear Filter
Remove Empty
Text Filters

Search

☒ (Select All)
☒ Manager
☒ Partner
☒ Senior
☒ Sr Manager
☒ Sr. Manager
☒ Staff

OK Cancel

(B) JobTitle Column: Frequencies

Column statistics		Value distribution	
Count	24	Senior	
Error	0	Staff	
Empty	0	Manager	
Distinct	6	Partner	
Unique	1	Sr. Manager	
Empty string	0	Sr Manager	
Min	Manag...		
Max	Staff		

5.4 Áp dụng mẫu để Chuyển đổi Cột (Column Transformation)

- **1. Làm sạch (Cleaning):** Xóa khoảng trắng thừa (Trim), sửa lỗi chính tả.
- **2. Định dạng (Formatting):** Đổi định dạng tiền tệ (VND sang USD), định dạng ngày tháng (MM/DD/YYYY sang DD/MM/YYYY).
- **3. Tách cột (Splitting):** Tách cột "Họ và Tên" thành hai cột "Họ" và "Tên".
- **4. Ghép cột (Concatenating):** Nối "Mã vùng" và "Số điện thoại".
- **5. Rút trích (Extracting):** Lấy tên miền từ địa chỉ Email.
- **6. Tính toán (Calculated):** Cột "Thành tiền" - $\text{"Số lượng"} \times \text{"Đơn giá"}$.

ILLUSTRATION 5.26

Current Value	New Value
MSU	Michigan State University
UM	University of Michigan

ILLUSTRATION 5.27

ABC MaritalStatus	
Married	
Single	
Married	
Single	
	null
Married	
	null
	null
Married	
Single	
	null

MaritalStatus

16 (67%)	0 (0%)	8 (33%)
Valid	Error	Empty

💡 Remove Empty ...

ILLUSTRATION 5.28

Description	Sample Code
The actual hours for a service must be positive and can't exceed 14.	<pre>ACTUALHOURSVALID = IF SERVICE.ACTUALTIME > 0 AND SERVICEACTUALTIME <= 14, THEN "YES", ELSE "NO"</pre>
The minimum employee rate is \$150, and the maximum employee rate must be lower than \$500.	<pre>RATEVALID = IF EMPLOYEE.RATE >= 150 AND EMPLOYEE.RATE < 500, THEN "YES", ELSE "NO"</pre>
Valid job titles are: {Manager, Partner, Senior, Sr. Manager, Staff}.	<pre>JOBTITLEVALID = IF EMPLOYEE.JOBTITLE IN {"MANAGER", "PARTNER", "SENIOR", "SR. MANAGER", "STAFF"}, THEN "YES", ELSE "NO"</pre>

ILLUSTRATION 5.29

Column Statistics	...
Count	4,632
Error	0
Empty	0
Distinct	4,632
Unique	4,632
NaN	0
Zero	0
Min	1
Max	4,632
Average	2,316.5
Standard deviation	1,337.28...
Even	2,316
Odd	2,316

ILLUSTRATION 5.30

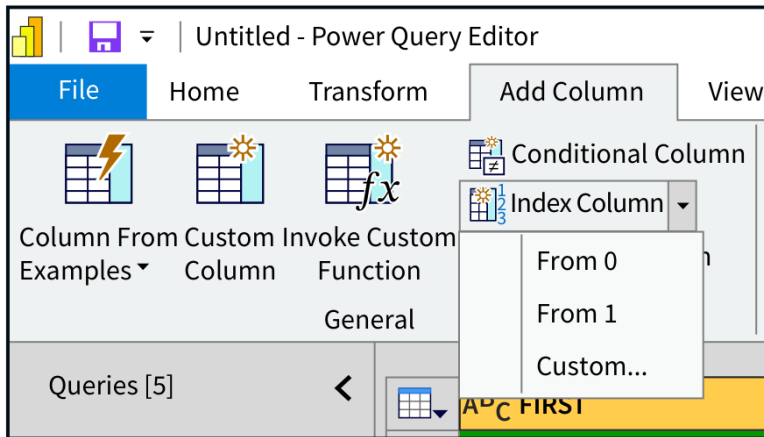


ILLUSTRATION 5.32

**(A) Monthly Sales Data
Across Tables**

JanuarySales

Number	Date	Amount
189	1/10/2025	\$ 17,450
190	1/15/2025	\$ 23,890
191	1/24/2025	\$ 19,001

FebruarySales

Number	Date	Amount
192	2/9/2025	\$ 25,451
193	2/10/2025	\$ 34,881
194	2/14/2025	\$ 7,282
195	2/23/2025	\$ 13,209

**(C) Monthly Sales Data Union
Sales**

Number	Date	Amount
189	1/10/2025	\$ 17,450
190	1/15/2025	\$ 23,890
191	1/24/2025	\$ 19,001
192	2/9/2025	\$ 25,451
193	2/10/2025	\$ 34,881
194	2/14/2025	\$ 7,282
195	2/23/2025	\$ 13,209

**(B) Product Information
Across Tables**

ProductDescriptions

ID	Type	Category
1	Stuffed Animals	Toys
2	Radio	Electronics
3	Soccer Balls	Toys

ProductAccounting

ID	QOH
1	84
2	133
3	354

(D) Product Information Merge

Product

ID	Type	Category	QOH
1	Stuffed Animals	Toys	84
2	Radio	Electronic	133
3	Soccer Balls	Toys	354

ILLUSTRATION 5.33

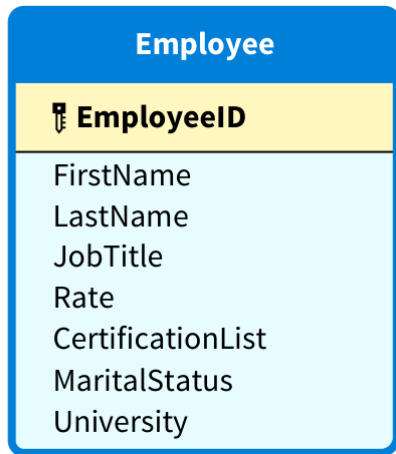


ILLUSTRATION 5.34

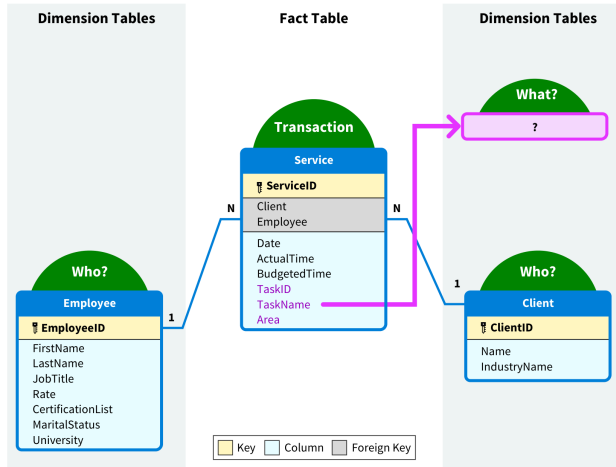
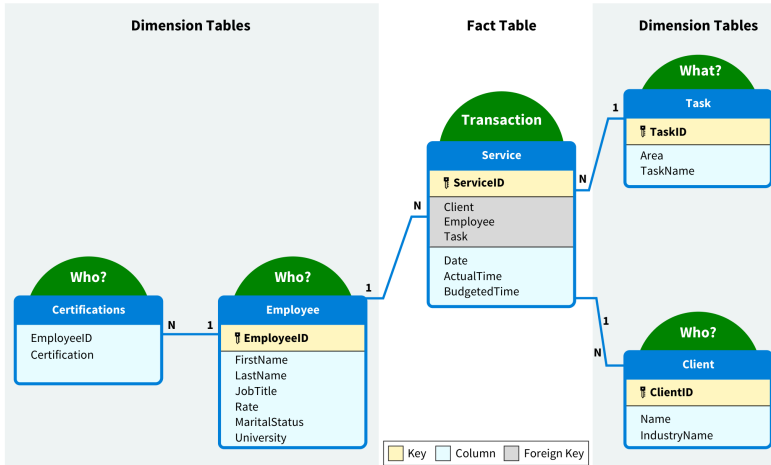


ILLUSTRATION 5.35



5.5 Áp dụng mẫu để Chuyển đổi Bảng (Table Transformation)

- **1. Lọc (Filtering):** Giữ lại các hàng thỏa mãn điều kiện (Ví dụ: Chỉ lấy doanh thu > 1000 USD).
- **2. Sắp xếp (Sorting):** Sắp xếp bảng theo thứ tự (Ví dụ: Bán chạy nhất lên đầu).
- **3. Gộp bảng theo cột (Join/Merge):** Kết hợp bảng "Hóa đơn" và "Khách hàng" dựa trên khóa chung (Mã KH).
- **4. Nối bảng theo hàng (Append/Union):** Nối dữ liệu bán hàng tháng 1 và tháng 2 vào chung một bảng (Phải có cùng số lượng cột và kiểu dữ liệu).

ILLUSTRATION 5.36

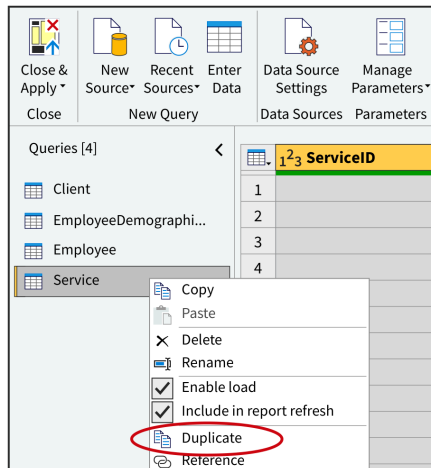


ILLUSTRATION 5.37

	1^2_3 TaskID	A^B_C Area	A^B_C TaskName
1	1	Accounting	Prep
2	2	Accounting	Review
3	3	Accounting	Admin
4	4	Accounting	Other
5	5	Tax	Prep
6	6	Tax	Review
7	7	Tax	Admin
8	8	Tax	Other

ILLUSTRATION 5.38

The screenshot shows the Microsoft Excel ribbon with the 'Transform' tab selected. The 'Unpivot Columns' dropdown menu is open, and 'Unpivot Other Columns' is circled in red. Below the ribbon, a table with 7 rows and 6 columns is shown. The first column is 'EmployeeID' (highlighted in yellow), and the other five columns are 'AB_CertificationList.1' through 'AB_CertificationList.4'. The data in the table is as follows:

	EmployeeID	AB_CertificationList.1	AB_CertificationList.2	AB_CertificationList.3	AB_CertificationList.4
1	1	null	null	null	null
2	2	CPA	null	null	null
3	3	CPA	CMA	null	null
4	4	null	null	null	null
5	5	null	null	null	null
6	6	CPA	null	null	null
7	7	CPA	CMA	CFA	null

ILLUSTRATION 5.39

Description	Sample Code
Only managers, sr. managers, and partners can review an engagement.	<pre>NOAUTHORITY = IF TASK.TASKNAME = "REVIEW" AND EMPLOYEE[JOBTITLE] IN {"MANAGER", "SR. MANAGER", "PARTNER"}, THEN "OK", ELSE "ISSUE"</pre>

ILLUSTRATION 5.40

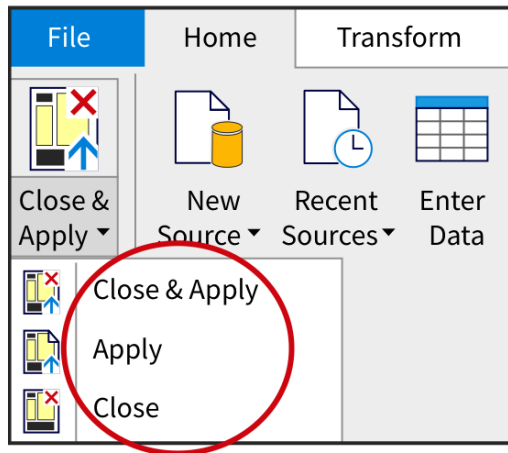


ILLUSTRATION 5.41

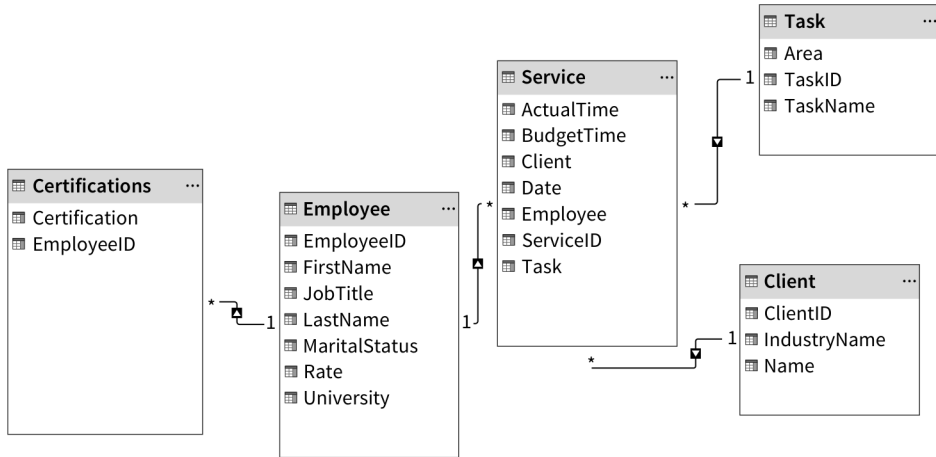


ILLUSTRATION 5.42

Edit Relationship ×

Select tables and columns that are related.

Certifications ▼

EmployeeID	Certification
2	CPA
3	CPA
3	CMA

Employee ▼

EmployeeID	JobTitle	Rate	MaritalStatus	FirstName	LastName	University
1	Staff	165	Married	Kayne	James	Michigan State University
2	Manager	250	Single	Ivan	Lenk	University of Michigan
3	Senior	225		Mick	Richards	Wayne State University

Fields for relationship definition

Cardinality specification

Cardinality ▼ Cross filter direction ▼

Navigation direction

Many to one (*:1) Single

☒ Make this relationship active ☐ Apply security filter in both directions

☐ Assume referential integrity

OK Cancel

ILLUSTRATION 5.42_1

Manage Relationships

Active	From: Table (Column)	To: Table (Column)
☒	Certifications (EmployeeID)	Employee (EmployeeID)
☒	Service (Client)	Client (ClientID)
☒	Service (Employee)	Employee (EmployeeID)
☒	Service (Task)	Task (TaskID)

New...

Autodetect...

Edit...

Delete

Close

ILLUSTRATION 5.43

Edit Relationship ×

Select tables and columns that are related.

Certifications ▼

EmployeeID	Certification
2	CPA
3	CPA
3	CMA

Employee ▼

EmployeeID	JobTitle	Rate	MaritalStatus	FirstName	LastName	University
1	Staff	165	Married	Kayne	James	Michigan State University
2	Manager	250	Single	Ivan	Lenk	University of Michigan
3	Senior	225		Mick	Richards	Wayne State University

Fields for relationship definition

Cardinality specification

Cardinality: Many to one (*:1) ▼

Cross filter direction: Single ▼

Navigation direction

☒ Make this relationship active ☐ Apply security filter in both directions

☐ Assume referential integrity

OK Cancel

5.6 Áp dụng mẫu để Chuyển đổi Mô hình (Model Transformation)

- **Tổ chức Schema:** Từ nhiều bảng rời rạc, liên kết chúng lại thành một mô hình thống nhất để phân tích.
- **Star Schema (Mô hình Ngôi sao):** Gồm 1 Bảng Sự kiện (Fact Table) ở giữa chứa dữ liệu giao dịch, và các Bảng Danh mục (Dimension Tables) xung quanh. Tối ưu cho tốc độ truy vấn.
- **Snowflake Schema (Mô hình Bông tuyết):** Giống Star Schema nhưng các Bảng Danh mục được chuẩn hóa (chia nhỏ) tiếp. Tiết kiệm không gian lưu trữ nhưng làm chậm tốc độ phân tích.

ILLUSTRATION 5.43_1

Edit Relationship ×

Select tables and columns that are related.

Certifications ▼

EmployeeID	Certification
2	CPA
3	CPA
3	CMA

Fields for relationship definition

Employee ▼

EmployeeID	JobTitle	Rate	MaritalStatus	FirstName	LastName	University
1	Staff	165	Married	Kayne	James	Michigan State University
2	Manager	250	Single	Ivan	Lenk	University of Michigan
3	Senior	225		Mick	Richards	Wayne State University

Cardinality specification

Cardinality ▼ Cross filter direction ▼

Many to one (*:1) Single

☒ Make this relationship active ☐ Apply security filter in both directions

☐ Assume referential integrity

OK Cancel

Navigation direction

ILLUSTRATION 5.44

Service				
Name	Description	Data Type	Key	Mandatory
Client	A client's unique ID.	Integer	Foreign	Yes
Employee	An employee's unique ID.	Integer	Foreign	Yes
Task	A task's unique ID.	Integer	Foreign	Yes
ServiceID	A service's unique ID.	Integer	Primary	Yes
Date	The date the service occurred.	Date		Yes
ActualTime	The actual time spent on the service provided.	Decimal		Yes
BudgetedTime	The budgeted time for the service provided.	Decimal		Yes
Task				
Name	Description	Data Type	Key	Mandatory
TaskID	A task's unique ID.	Integer	Primary	Yes
TaskName	A task's name.	Text		Yes
Area	A task's area.	Text		Yes
Client				
Name	Description	Data Type	Key	Mandatory
ClientID	A client's unique ID.	Integer	Primary	Yes
Name	A client's name.	Text		Yes
IndustryName	A client's industry.	Text		Yes
Employee				
Name	Description	Data Type	Key	Mandatory
EmployeeID	An employee's unique ID.	Integer	Primary	Yes
FirstName	An employee's first name.	Text		Yes
LastName	An employee's last name.	Text		Yes
JobTitle	An employee's job title.	Text		Yes
Rate	The hourly rate charged for the employee.	Integer		Yes
MaritalStatus	An employee's marital status.	Text		No
University	The university from which the employee received an undergraduate degree.	Text		No
Certifications				
Name	Description	Data Type	Key	Mandatory
EmployeeID	An employee's unique ID.	Integer	Foreign	Yes
Certification	A certification earned by the employee.	Text		Yes

ILLUSTRATION 5.45

Data Preparation Patterns				
Extraction				
Id	Code	Issue	Detect (Data Profiling)	Correct (ETL)
1	V	All the data did not transfer.	Compare row counts.	Add missing rows.
2	V	Data was not correctly transferred.	Compare control amounts.	Modify incorrect values.
Transformation				
Id	Code	Issue	Detect (Data Profiling)	Correct (ETL)
Column				
3	S	Irrelevant or unreliable data.	Visually scan columns for irrelevant and unreliable data.	Remove columns with irrelevant and unreliable data.
4	S	Incorrect or ambiguous data.	Scan columns for incorrect or ambiguous names.	Rename columns.
5	S	Incorrect data types.	Inspect the data type.	Change data type.
6	S	Columns are not single-valued.	Examine the values in the columns.	Split composite columns. Create a separate table for multi-valued columns.
7	V	Incorrect values.	Detect incorrect values with outliers.	Modify incorrect values.
8	V	Inconsistent values.	Identify inconsistent values.	Modify inconsistent values.
9	V	Incomplete values.	Investigate null values.	Remove the columns or replace the null values.
10	V	Invalid values.	Create and apply validation rules.	Modify invalid values.
Table				
11	S	Incorrect or ambiguous table names.	Visually scan the table for incorrect or ambiguous names.	Rename tables.
12	S	Missing primary keys.	Identify tables missing a primary key.	Create a primary key.
13	S	Redundant column content.	Perform column-by-column comparisons.	Delete redundant and dependent data.
14	V	Invalid values detected with intra-table rules.	Create and apply intra-table validation rules.	Modify the invalid data.
Model				
15	S	Data spread across tables.	Identify tables with a similar structure or tables that describe different characteristics of the same entity.	Union or merge tables.
16	S	Compliance with dimensional modeling principles.	Analyze a data model's compliance with dimensional modeling principles.	Restructure model as a star/snowflake schema.
17	V	Invalid values detected with inter-table rules.	Create and apply inter-table validation rules.	Modify the invalid rules.
Loading				
Id	Code	Issue	Detect (Data Profiling)	Correct (ETL)
18	V	Incomplete data loading.	Compare row counts.	Add the missing data.
19	V	Incorrect data loading.	Compare control amounts.	Modify incorrect values.
20	S	Missing or incorrect data relationships.	Investigate the completeness and accuracy of the data model.	Create, modify, and delete relationships. Modify the data model.

ILLUSTRATION 5.46

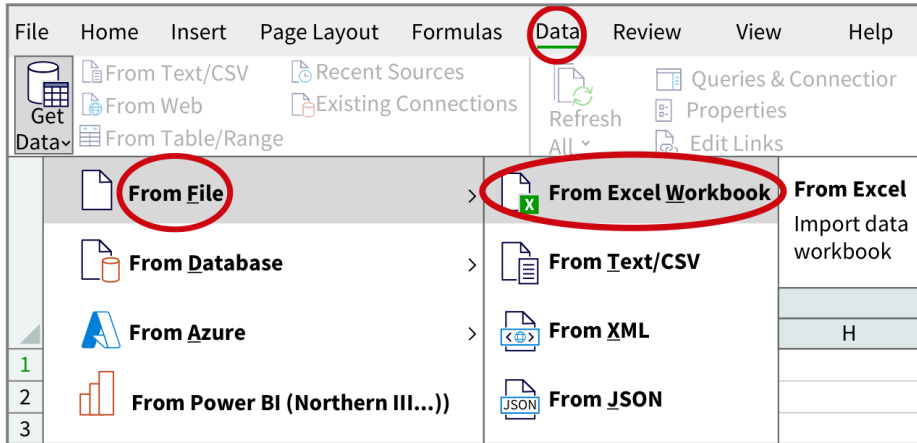


ILLUSTRATION 5.47

Navigator

☐ Select multiple items

Display Options ▾

- Beans (1).xlsx [4]
 - ClData
 - E-Dem
 - Employee
 - Service**

Service

ID	Date	ActualTime	BudgetedTime	ID_1	Area
1	1/1/2025	3.75	3.5	2	Accountin
2	1/2/2025	2.65	2.5	1	Accountin
3	1/2/2025	0.07	0.1	1	Accountin
4	1/2/2025	0.35	0.3	1	Accountin
5	1/2/2025	0.1	0.1	1	Accountin
6	1/2/2025	0.35	0.4	1	Accountin
7	1/2/2025	0.07	0.1	1	Accountin
8	1/2/2025	0.25	0.3	1	Accountin
9	1/2/2025	0.25	0.2	1	Accountin

Load ▾ Transform Data Cancel

ILLUSTRATION 5.48

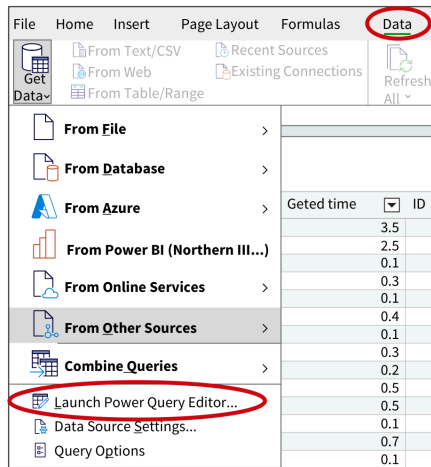


ILLUSTRATION 5.49

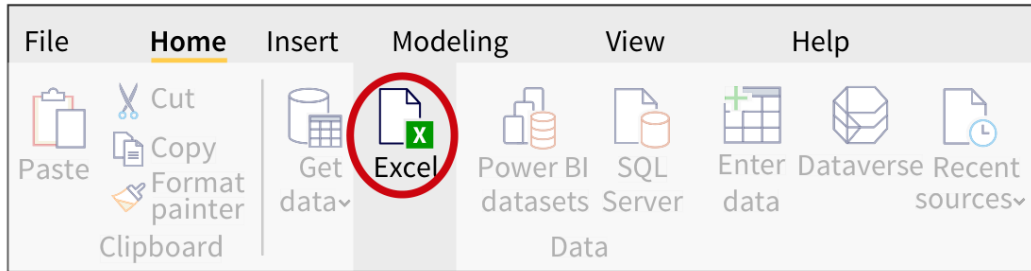


ILLUSTRATION 5.50

Navigator

Display Options ▾

Beans.xlsx [4]

☐ ClData

☐ E-Dem

☐ Employee

☒ Service

Service

Preview downloaded on Wednesday, March 9, 2022

ID	Date	ActualTime	BudgetedTime	ID_1	Area
1	1/1/2025	3.75	3.5	2	Accountin
2	1/2/2025	2.65	2.5	1	Accountin
3	1/2/2025	0.07	0.1	1	Accountin
4	1/2/2025	0.35	0.3	1	Accountin
5	1/2/2025	0.1	0.1	1	Accountin
6	1/2/2025	0.35	0.4	1	Accountin
7	1/2/2025	0.07	0.1	1	Accountin
8	1/2/2025	0.25	0.3	1	Accountin
9	1/2/2025	0.25	0.2	1	Accountin
10	1/2/2025	0.75	0.5	2	Accountin
11	1/2/2025	0.75	0.5	8	Tax
12	1/2/2025	0.1	0.1	1	Accountin
13	1/2/2025	0.75	0.7	8	Tax
14	1/2/2025	0.1	0.1	1	Accountin
15	1/2/2025	0.25	0.2	1	Accountin
16	1/2/2025	0.07	0.1	1	Accountin
17	1/2/2025	0.35	0.4	1	Accountin
18	1/2/2025	0.07	0.1	1	Accountin
19	1/2/2025	0.07	0.1	1	Accountin
20	1/2/2025	0.35	0.5	1	Accountin
21	1/2/2025	3.95	3.8	4	Accountin
22	1/2/2025	4	2.8	2	Accountin

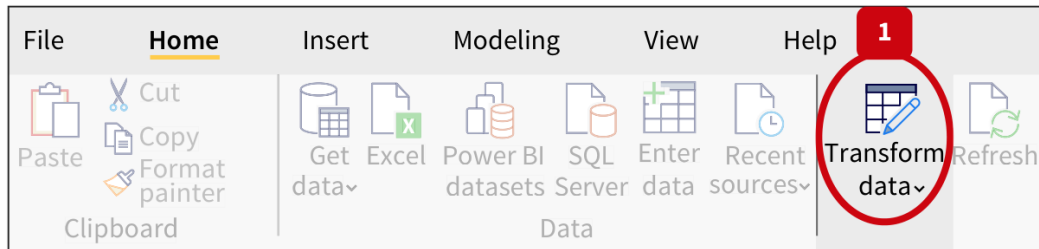
Load

Transform Data

Cancel

ILLUSTRATION 5.51

Power BI



5.7 Áp dụng mẫu cho vấn đề Tải Dữ liệu (Data Loading)

- **Tải ban đầu (Initial Load):** Nạp toàn bộ dữ liệu lịch sử vào một Data Warehouse mới tinh. Thường tốn nhiều thời gian.
- **Tải tăng dần (Incremental Load):** Chỉ nạp những giao dịch mới phát sinh, chạy định kỳ hàng ngày/hàng giờ.
- **Tải làm mới (Full Refresh):** Xóa toàn bộ dữ liệu cũ và nạp lại toàn bộ dữ liệu mới. Chỉ dùng cho các bảng nhỏ và ít thay đổi.
- **Kiểm tra tính toàn vẹn (Integrity Checks):** Phải đảm bảo số dòng của bảng Nguồn (Source) khớp với bảng Đích (Destination) sau khi tải.

ILLUSTRATION 5.52

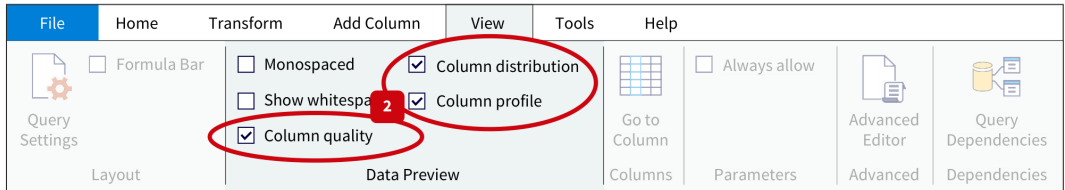


ILLUSTRATION 5.53

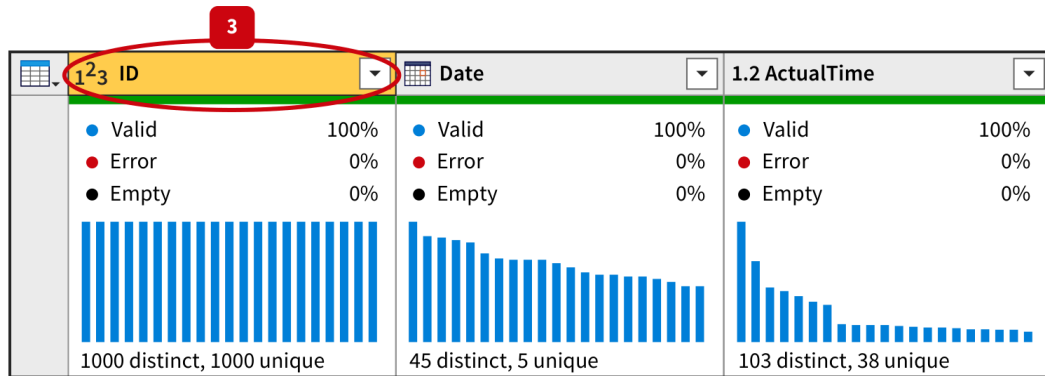


ILLUSTRATION 5.54

10 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

ILLUSTRATION 5.55



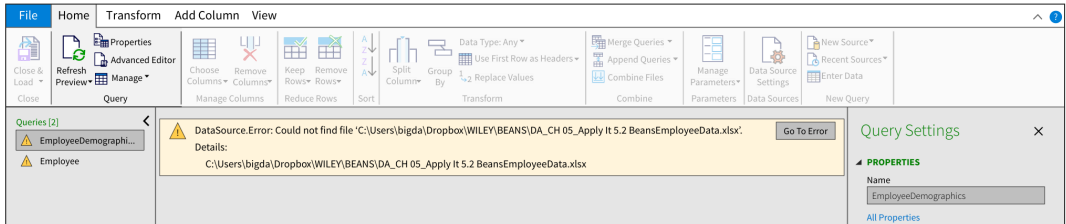
Column profiling based on top 1000 rows

Column profiling based on entire data set

ILLUSTRATION 5.56

Column statistics	...
Count	4,632
Error	0
Empty	0
Distinct	4,632
Unique	4,632
NaN	0
Zero	0
Min	1
Max	4,632
Average	2,316.5
Standard deviation	1,337.28...
Even	2,316
Odd	2,316

ILLUSTRATION 5.57



Apply It

ILLUSTRATION 5.58

Merge

Select a table and matching columns to create a merged table.

Employee

EmployeeID	FirstName 1	LastName 2	JobTitle	Rate	MaritalStatus
1	Kayne	James	Staff	165	Married
2	Ivan	Lenk	Manager	250	Single
3	Mick	Richards	Senior	225	
4	Juan	Lozano	Staff	180	Married
5	Gail	David	Senior	218	Single

EmployeeDemographics

FirstName 1	LastName 2	CertificationList	University
Diego	Asare	null	Grand Valley State University
Kim	Bush	CPA	University of Delaware
Marcus	Clay	null	Michigan State University
Gail	David	null	University of Michigan
Shiela	Diamond	CPA, CMA, CFA	Michigan State University

Join Kind

Left Outer (all from first, matching from second)

☒ Use fuzzy matching to perform the merge

Fuzzy matching options

Similarity threshold (optional)

0.4

☒ Ignore case

☒ Match by combining text parts

OK Cancel

ILLUSTRATION 5.59

	EmployeeDemographics	
Table		6
Table		
Table		
Table		
Table		

ILLUSTRATION 5.60

The screenshot shows a software interface with a list of queries on the left and a data table on the right. A red circle highlights the 'EmployeeDemographics' query in the list, with a red square containing the number '7' next to it. A context menu is open over the 'EmployeeDemographics' query, displaying options: Copy, Paste, Delete, Rename, Enable load, and Include in report refresh (which is checked).

Queries [4]

- Client
- EmployeeDemographics**
- Employee
- Service

	FirstName
1	Kayne
2	Ivan

- Copy
- Paste
- Delete
- Rename
- Enable load
- ☒ Include in report refresh

ILLUSTRATION 5.61

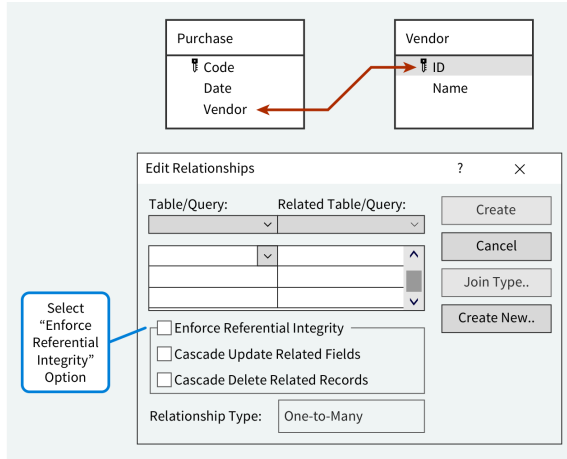


ILLUSTRATION 5.62

(A)

ID ▼	Name ▼
1	Billy
2	Namsun
3	Esmeralda
4	Cassius
5	Carol

(B)

Code ▼	Date ▼	Vendor ▼	Click to Add ▼
L122	1/2/2023	1	
U18	1/3/2023	6	

Microsoft Access

 You cannot add or change a record because a related record is required in table 'VENDOR'.

OK Help

Entering a Non-Existing Vendor

Error Message
Generated By
Access

Apply It 5.1

Code	Description	QOH
BASHL	Bashful Large, Dwarfs	13
BASHS	Bashful Small, Dwarfs	67
BERL	Berlioz, Aristocats	23
CAPH	Captain Haddock, Tintin	14
DOCL	Doc Large, Dwarfs	23
DOCS	Doc Small, Dwarfs	45
DOPL	Dopey Large, Dwarfs	43
DOPS	Dopey Small, Dwarfs	-3
DUCH	Duchess, Aristocrats	44

Apply It 5.2

Aristocats		
Code	Description	QOH
BERL	Berlioz	23
DUCH	Duchess	44
MAR	Marie	34
THO	Thomas	19
TOU	Toulouse	32

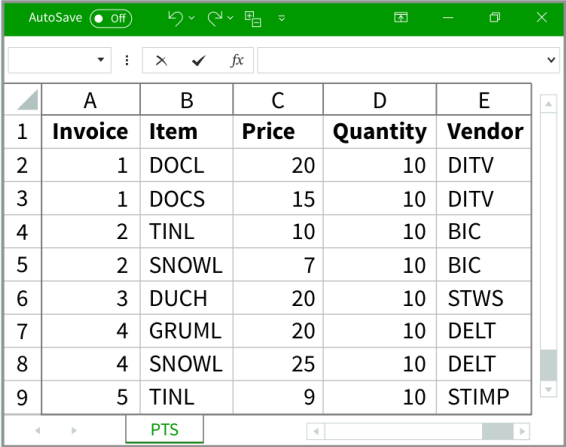
Dwarfs	
Code	Description
BASHL	Bashful Large
BASHS	Bashful Small
DOCL	Doc Large
DOCS	Doc Small
DOPL	Dopey Large
DOPS	Dopey Small
GRUML	Grumpy Large
GRUMS	Grumpy Small
HAPL	Happy Large
HAPS	Happy Small
SLB	Sleepy Big
SLS	Sleepy Small
SNEL	Sneezy Large
SNES	Sneezy Small
SNOWL	Snow White Large
SNWS	Snow White Small

Tintin		
Code	Description	QOH
CAPH	Captain Haddock	14
SNO	Snowy	22
TINL	Tintin Large	23
TINS	Tintin Small	44

Apply It 5.3

Column Statistics		...
Count		28
Error		0
Empty		0
Distinct		8
Unique		4
NaN		0
Zero		0
Min		10
Max		40
Average		25.3214...

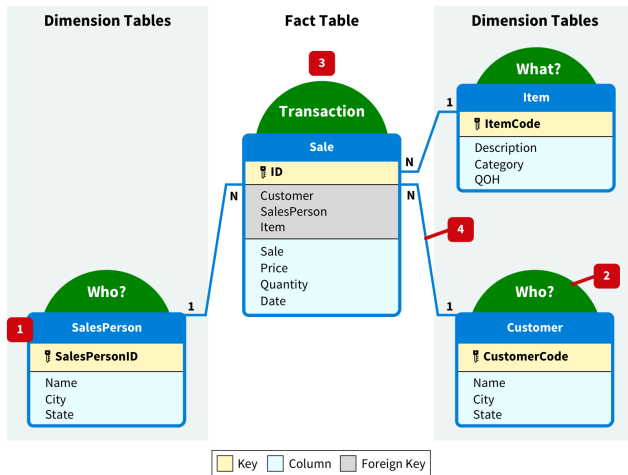
Apply It 5.5



The screenshot shows a Microsoft Excel window with a green title bar. The 'AutoSave' toggle is set to 'Off'. The formula bar is empty. The spreadsheet contains a table with 6 columns: Invoice, Item, Price, Quantity, and Vendor. The data is organized into 9 rows, with the first row serving as the header. The 'Invoice' column contains values 1, 1, 2, 2, 3, 4, 4, and 5. The 'Item' column contains DOCL, DOCS, TINL, SNOWL, DUCH, GRUML, SNOWL, and TINL. The 'Price' column contains 20, 15, 10, 7, 20, 20, 25, and 9. The 'Quantity' column contains 10, 10, 10, 10, 10, 10, 10, and 10. The 'Vendor' column contains DITV, DITV, BIC, BIC, STWS, DELT, DELT, and STIMP. The status bar at the bottom indicates the active cell is 'PTS'.

	A	B	C	D	E
1	Invoice	Item	Price	Quantity	Vendor
2	1	DOCL	20	10	DITV
3	1	DOCS	15	10	DITV
4	2	TINL	10	10	BIC
5	2	SNOWL	7	10	BIC
6	3	DUCH	20	10	STWS
7	4	GRUML	20	10	DELT
8	4	SNOWL	25	10	DELT
9	5	TINL	9	10	STIMP

Apply It 5.6



Apply It 5.7

The screenshot displays the Power BI Desktop interface. At the top, two tables are visible: 'Customer' and 'Sale'. The 'Customer' table has columns: City, CustomerCode, Name, and State. The 'Sale' table has columns: Customer, Date, ID, Item, Price, Quantity, Sale, and SalesPerson. A relationship line connects the 'CustomerCode' column in the 'Customer' table to the 'Customer' column in the 'Sale' table, with a cardinality of 1 to 1.

Below the tables, the 'Edit Relationship' dialog box is open. It shows the selected tables and columns for the relationship. The 'Customer' table is selected, and the 'Sale' table is also selected. The dialog displays the following data:

CustomerCode	Name	City	State
1	Cruella De Vil	Phoenix	AZ
2	Scar LeRoI	Orlando	FL
3	Winnie Pooh	Wilmington	DE

ID	Sale	Item	Price	Quantity	Customer	SalesPerson	Date
1	1	DOCL	30	10	1	111223333	Friday, February 21, 2025
2	1	GRUML	30	10	1	111223333	Friday, February 21, 2025
3	2	SNES	20	20	6	null	Saturday, February 22, 2025

The dialog also shows the 'Cardinality' set to 'One to one (1:1)' and the 'Cross filter direction' set to 'Both'. The 'Make this relationship active' checkbox is checked. The 'Assume referential integrity' checkbox is unchecked. The 'OK' and 'Cancel' buttons are at the bottom right.

Brief Exercises

BE 5.1 & BE 5.2

a. Dimension tables

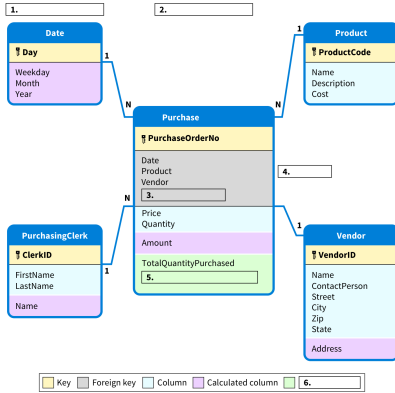
b. Fact table

c. N

d. Measure

e. TotalAmountPurchased

f. PurchasingClerk



1

Customer

1

N

Sales

2

Vendor

1

N

Purchases

3

Employee

1

N

Order

4

Employee

1

N

Payment

BE 5.3 & BE 5.4

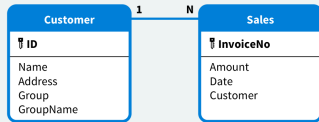
CustomerInfo	Gender	Date	Email
Steve Millier, 121 South Beach St	M	10/21/1978	steve.miller
Sara Stevens, 435 Parker St	Female	9/21/2008	sara.stevens@mail.com
Libby Ralston, 812 Foster Ave	Female	6/28/2956	libby123@mail.com
Able Meyers, 902	Male	1/4/2009	smileyone@mail.com
Spring Stevens, 639 Bulldog Ct	Undisclosed	6/14/1982	sparkie.steven
Kevin Hogan, 3098 N. Tier St. Apt 392	M	7/22/1955	doglover@mail.com
Libby Ralston, 812 Foster Ave	Female		
Sara Stevens, 435 Parker St	Female	9/21/2008	sara.stevens@mail.com
Cooper Mazzu, 2342 Lincoln Way	Male	4/17/1955	coop8623
Molly White	F	10/21/1977	molly.white.123@mail.com
Janet Jones, 909 Hwy 78	F		

Table A		
Description	QOH	Location
Blue Dog Collar	5	A10
Red Doc Collar	3	A11
7 Inch Rope Sturdy Lead	10	B10
10 Inch Rope Sturdy Lead - Green	3	B10
10 Inch Rope Sturdy Lead - Black	5	B10

Table B			
ProductDescription	Cost	Price	Sale
Dog Collar - Blue	12.54	16.93	Y
7 Inch Rope Sturdy Lead	15.44	20.85	N
Red Dog Collar	15.54	20.98	N
10 Inch Rope Sturdy Lead - Bk	20.38	27.52	N
Green 10 Inch Rope Sturdy Lead	28.34	38.26	Y

BE 5.5 & BE 5.6

(A)



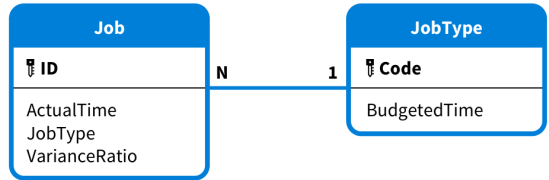
(B)



SaleID	Date	SaleLocation	SaleAmount	LocalTax	SalesTax	VAT
1001	5/25/2024	USA	\$ 197	0.02	0.0685	
1002	5/26/2024	USA	\$ 113	0.02	0.0685	
1003	5/27/2024	MEX	\$ 155			0.16
1004	5/28/2024	MEX	\$ 423			0.16
1005	5/29/2024	CAN	\$ 88			0.05
1006	5/30/2024	AUS	\$ 250			0.1

BE 5.13 & BE 5.15

	ABC LastName	ABC FirstName	ABC Location	ABC Gender	123 Level	123 2024Hours	123 RateperHour
1	James	Kayne	Houston	Male	1	3,521	27
2	Lenk	Ivan	Texarcane	Male	3	3,811	35
3	Richards	Mick	Chicago	Male	5	2,901	39
4	Lozano	Juan	Boston	Male	4	3,477	37
5	David	Gail	Chicago	Female	7	3,420	40
6	Petroni	Ann	Houston	Female	1	1,709	27
7	Diamond	Shiela	Boston	Female	3	2,331	35
8	Sutherland	Yvonne	Chicago	Female	4	3,871	37
9	Clay	Marcus	Boston	Male	5	1,942	39
10	McCarthy	Molly	Chicago	Female	1	3,665	27

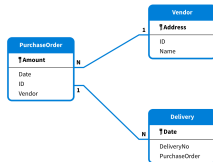


PurchaseOrder (Information about orders placed)	
Name	Description
Amount	The dollar amount to be paid to the vendor.
Date	The date on which the order was placed.
ID	The purchase order's unique ID.
Vendor	The ID of the vendor from whom the goods were purchased.

Delivery (Information about goods delivered)	
Name	Description
Date	The date on which the goods were delivered.
DeliveryNo	The delivery's unique ID.
PurchaseOrder	The purchase order number associated with the delivery.

Vendor (Information about vendors)	
Name	Description
Address	The vendor's address.
ID	The vendor's unique ID.
Name	The vendor's name.

Data Model:



Relationships:

Manage Relationships		
Active	From: Table (Column)	To: Table (Column)
☒	Delivery (DeliveryNo)	PurchaseOrder (ID)
☒	PurchaseOrder (Vendor)	Vendor (Name)

Exercises

EX 5.2 & EX 5.3

BargainVendors

	ABC ID	ABC Name	ABC Email
1	US1	Cohen, Sandra	scohen@vstores.com
2	US2	Gray, Sev	sev-gray@ibox.com
3	US3	Leroy, Shanice	sleroy@google.com
4	US4	Morales, Benita	bmor@hf.com
5	US5	Wang, CeCe	cece@cch.com
6	CAN1	Zhu, Jim	jzhu@outlook.com
7	CAN2	Coyle, Irene	irene@coyle.com
8	CAN3	Dabrowski, Elizbieta	ElzDab@gmail.com
9	CAN4	King, Laura	LauraKing@fmh.com
10	CAN5	Jackson, Kiara	kiaraj@ncn.com

BargainPurchases

	123 LineItemID	123 OrderNo	Date	ABC Type	ABC Category	123 Price	123 Quantity	ABC Vendor
1	1	8992	1/2/2025	Rice Cooker	Kitchen	\$ 69	2	US1
2	2	8993	1/2/2025	Phone	Electronics	\$ 113	1	CAN3
3	3	8994	1/2/2025	Fryer	Kitchen	\$ 181	1	CAN5
4	4	8894	1/2/2025	Stuffed Animal	Toys	\$ 14	5	US2
5	5	8896	1/2/2025	TV	Electro	\$ 79	1	CAN2
6	6	8897	1/3/2025	Stuffed Animal	Toys	\$ 67	1	CAN1
7	7	8897	1/3/2025	Microwave	Kitchen	\$ 205	1	US1
8	8	8898	1/3/2025	Headphone	Electronics	\$ 45	1	US4
9	9	8898	1/3/2025	Jewelry	Toys	\$ 13	2	CAN1
10	10	8899	1/3/2025	Printer	Electro	\$ 199	2	CAN3

A

SalesInformation

1,1/6/2025,97,0,0

2,1/6/2025,113,2,3,49

3,1/6/2025,55,0,1,23

4,1/7/2025,423,2,0

5,1/7/2025,88,0,0

6,1/7/2025,250,3,5,53

7,1/8/2025,680,2,0

8,1/8/2025,58,0,0

9,1/8/2025,99,1,1

10,1/8/2025,258,2,0

EX 5.6 & EX 5.7

ID	ProductCode	Price
3044	Accord-Black-2017-*13,14,15*	12350
3045	CRV-White-2019-*11,22,27*	16990
3046	Accord-Green-2012-*4,7*	9766
3047	Civic-White-2015-*4,5,6*	10889
3048	Pilot-Red-2018-*1,7,11,18*	22998
3049	Civic-Gray-2017-*2,6,9,20*	11150
3050	Odyssey-White-2020-*4,7,11,12,17,*	24551
3051	Ridgeline-Black-2012-*1,13*	12990
3052	Pilot-White-2016-*7*	17000
3053	Accord-Black-2018-*12,14,17*	13500

Region	2022:Q1	2022:Q2	2022:Q3	2022:Q4	2023:Q1	2023:Q2	2023:Q3	2023:Q4	2024:Q1	2024:Q2	2024:Q3	2024:Q4
US	77,265,889	57,176,758	57,748,525	78,811,207	97,265,889	71,976,758	80,613,969	94,347,912	107,467,822	78,451,510	89,434,721	102,094,431
EUROPE	53,761,998	47,848,178	51,197,551	53,224,378	65,534,311	58,325,537	67,074,367	67,500,340	73,666,129	54,512,935	54,512,935	72,929,468
ASIA	32,188,799	27,038,591	31,094,380	32,832,575	31,777,112	23,197,292	23,661,238	32,094,883	29,778,112	22,035,803	24,239,383	28,289,206

WilkinsonWeek1

ID	Employee	JobNo	WeekNo	Thursday: 1/2/2025	Friday: 1/3/2025	Saturday: 1/4/2025
3405	1	ERF007	1	4	8	4
3406	1	ERF008	1	4		4
3407	13	ERF007	1	8	8	8
3408	25	IR68	1	4	8	0

WilkinsonWeek2

ID	Employee	JobNo	WeekNo	Monday: 1/6/2025	Tuesday: 1/7/2025	Wednesday: 1/8/2025	Thursday: 1/9/2025	Friday: 1/10/2025	Saturday: 1/11/2025
4608	25	ERF008	2	8	8	8	8	8	4
4609	13	ERF008	2	8	8	4			4
4610	13	IR68	2			4	8	8	
4611	1	IR101	2	6	8	10	8	8	4

Professional Application Cases (PAC)

PAC 5.1 & PAC 5.2

Vendors

Code	Name	Address
1	Discount store	4326 Lochmere Lane, Groton, CT, 06340
2	Lebron Jordan	1177 Gore Street, Houston, TX, 77027
3	Duffy Warehouse	4457 Lonely Oak Drive, Mobile, AL, 36603
4	Bob Dylan	3764 Courtright Street, Fargo, ND, 58102
5	Segher's Fabrics	2170 Metz Lane, Camden, NJ, 08102

Employees

Code	Name	Address
1	Joan Waddington	Phillps Circle 12, Fargo, ND, 58102
2	Matt Anthony	1012 Catherine Drive, St. Thomas, ND, 58276
3	Elizabeth Petroni	2490 Findley Avenue, Hope, ND, 58046
4	Andrea Dylan	3764 Courtright Street, Fargo, ND, 58102
5	Jimmy John	4000 Findley Avenue, Minot, ND, 58701

Purchases

Number	Date	Amount	Employee
1	1/6/2025	\$ 4,176	1
2	1/7/2025	\$ 660	3
3	1/9/2025	\$ 1,680	5
4	1/14/2025	\$ 4,703	4
6	1/20/2025	\$ 3,108	2
7	1/21/2025	\$ 2,263	4
8	1/23/2025	\$ 1,597	6
8	1/23/2025	\$ 1,597	6
9	1/30/2025	\$ 3,433	2
10	1/31/2025	\$ 1,833	4

SalesOrders

OrderNo	Date	Amount	Payments
1	1/6/2025	\$ 838.46	1
2	1/7/2025	\$ 100.03	2
3	1/9/2025	\$ 245.67	3,10
4	1/13/2025	\$ 3,632.16	
5	1/15/2025	\$ 386.90	5
6	1/20/2025	\$ 753.15	4
7	1/20/2025	\$ 611.00	7,9,11
8	1/23/2025	\$ 63.00	
9	1/25/2025	\$ 1,496.72	
10	1/28/2025	\$ 194.80	11

CashReceipts

CashReceiptNo	Date	Amount	Type
1	1/6/2025	\$ 838.46	Cash
2	1/7/2025	\$ 100.3	Credit Card
3	1/9/2025	\$ 150	Check
4	1/20/2025	\$ 350	Credit Card
5	1/24/2025	\$ 386.9	Cash
6	1/27/2025	\$ 200	Check
7	1/27/2025	\$ 500	Check
8	1/30/2025	\$ 403.15	Check
9	1/30/2025	\$ 200	Check
10	1/31/2025	\$ 95.67	Check
11	1/31/2025	\$ 194.8	Check

SalesTaxRate

No	Price:100	StandardShipping:0	ExpressShipping:0	Packing:25	Design:50
1010	700	38	0	50	0
1011	280	0	154	125	225
1012	1,590	50	0	0	0
1013	906	57	0	25	0
1014	142	49	0	250	150
1015	1,728	44	0	125	0
1016	664	0	252	35	0
1017	208	66	0	35	0
1018	778	60	0	0	0
1019	296	62	0	10	0
1020	800	75	0	0	0

State	SalesTaxRate
Connecticut	6.35
Delaware	N/A
Hawaii	4.17
Kentucky	6
Maryland	6
Michigan	6
Rhode Island	7

SalesOrder

OrderNo	Amount	SalesTax	State
1	\$ 791	\$ 47.46	KY
2	\$ 88	\$ 12.03	RI
3	\$ 231	\$ 14.67	CT
4	\$ 3,488	\$ 144.16	RI
5	\$ 365	\$ 21.90	MD
6	\$ 723	\$ 30.15	HI
7	\$ 611	\$ 0.00	DE
8	\$ 55	\$ 8.00	MD
9	\$ 1,412	\$ 84.72	MI
10	\$ 187	\$ 7.80	HI

SalesOrder

OrderNo	Amount	SalesTax	State
1	\$ 791	\$ 47.46	KY
2	\$ 88	\$ 12.03	RI
3	\$ 231	\$ 14.67	CT
4	\$ 3,488	\$ 144.16	RI
5	\$ 365	\$ 21.90	MD
6	\$ 723	\$ 30.15	HI
7	\$ 611	\$ 0.00	DE
8	\$ 55	\$ 8.00	MD
9	\$ 1,412	\$ 84.72	MI
10	\$ 187	\$ 7.80	HI

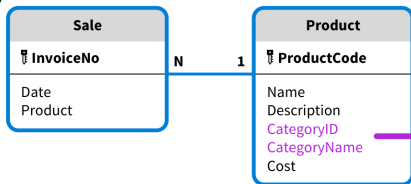
Tóm tắt Mục tiêu Học tập Chương 5

- **LO 5.1:** Data Profiling giúp xác định "sức khỏe" dữ liệu trước khi xử lý.
- **LO 5.2:** Nắm vững vai trò của từng bước trong quá trình ETL.
- **LO 5.3 & 5.7:** Sử dụng đúng mẫu Trích xuất (Full/Incremental) và mẫu Tải (Refresh/Incremental) để tối ưu hiệu suất.
- **LO 5.4 - 5.6:** Áp dụng thành thạo các kỹ thuật chuyển đổi Cột (Clean, Split), Bảng (Join, Append), và xây dựng Mô hình dữ liệu chuẩn (Star Schema).

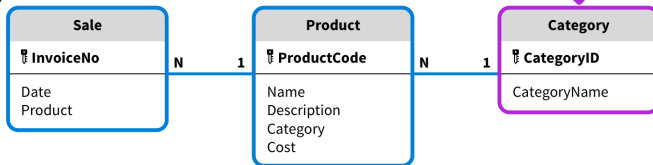
Applying critical thinking 5.3

Alternative Data Models for the Same Data Set

(A)



(B)



Dimension Tables

Date
 Date
WeekDay
Month
Year

Fact Table

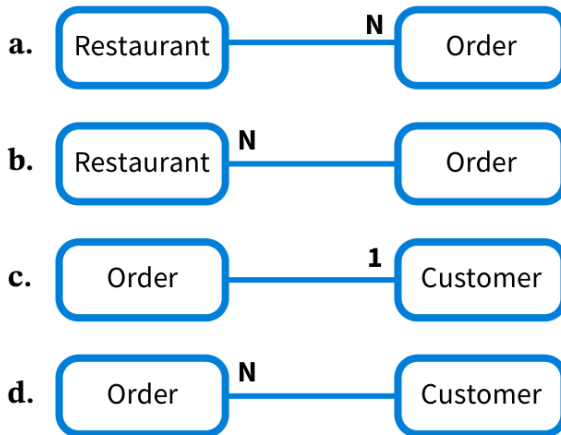
Purchase
 LineItemNumber
Employee
Item
PurchaseDate
Vendor
PurchaseID
Quantity
Price

Dimension Tables

Item
 ItemCode
Description

Employee
 EmployeeCode
Name
Position

Vendor
 VendorID
Name
Contacts



Hỏi & Đáp

Cảm ơn các bạn đã lắng nghe!